

REDUCE ITEMS AND ATTRIBUTES

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The Big Picture



Reducing Items and Attributes

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Reduce

Filter



Aggregate



Embed





Why Reduce?



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Attribute Aggregation:

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Reduction

Why and When to
Use DR?How to Show DR
Data?

Why Reduce

- Static data reduction idioms only reduce what is shown, as the name suggests. However, in the dynamic case, the outcome of changing a parameter or a choice may be an increase in the number of visible elements
 - **Reduction** includes *reduce* and *increase*
- Reducing the amount of data shown in a view is an obvious way to reduce its visual complexity
- Reduction can be applied to both *items* and *attributes*; the word **element** will be used to refer to either items or attributes when design choices that apply to both are discussed



Filter

- Item Filtering
- Attribute Filtering



Filter

Item Filtering

Attribute Filtering

Aggregate

Item Aggregation

Spatial Aggregation

Attribute Aggregation:

Dimensionality

Reduction

Why and When to
Use DR?

How to Show DR
Data?

Filter

- The design choice of **filtering** is a straightforward way to reduce the number of elements shown: some elements are simply eliminated
- Filtering can be applied to both items and attributes
- In an interactive vis context, filtering is often accomplished through **dynamic queries**



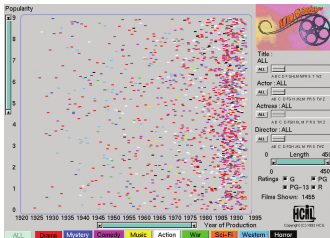
Item Filtering

- In **item filtering**, the goal is to eliminate items based on their values with respect to specific attributes.
- Fewer items are shown, but the number of attributes shown does not change.

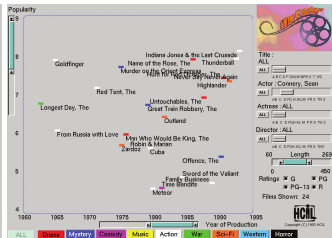


Example: FilmFinder

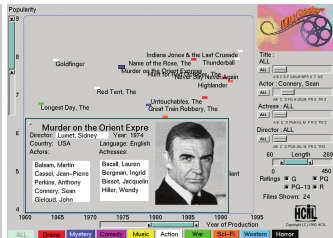
- FilmFinder features tightly coupled interactive filtering, where the result of moving sliders and pressing buttons is immediately reflected in the visual encoding.
- Exploration begins with an overview of all movies in the dataset.
 - Moving the actor slider to select Sean Connery filters out most of the other movies, leaving enough room to draw labels.
 - Clicking on the mark representing a movie brings up a detail view.



(a)



(b)



(c)

Idiom

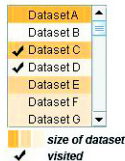
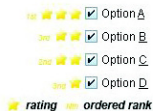
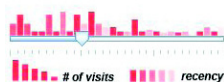


System	FilmFinder
What: Data	Table: nine value attributes.
How: Encode	Scatterplot; detail view with text/images.
How: Facet	Multiform, overview–detail.
How: Reduce	Item filtering.

Widgets



- Standard **widgets** for filtering controls can be augmented by concisely visually encoding information about the dataset, but in the part of the screen normally thought of as the control panel rather than a separate display area
- The scented widget idiom adds visual encoding information directly to standard graphical widgets to make filtering possible with high information density displays





Attribute Filtering

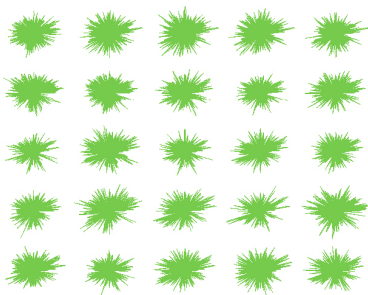
- With **attribute filtering**, the goal is to eliminate attributes rather than items; that is, to show the same number of items, but fewer attributes for each item.
- Item filtering and attribute filtering can be combined, with the result of showing both fewer items and fewer attributes.
- Attribute filtering is often used in conjunction with **attribute ordering**.



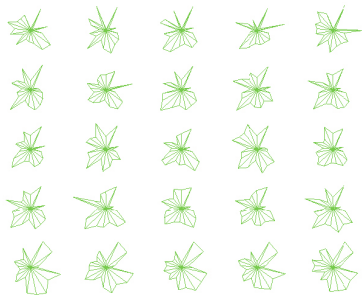
Example: DOSFA

The DOSFA idiom shown on star glyphs with a medical records dataset of 215 dimensions and 298 points.

- (a) The full dataset is so dense that patterns cannot be seen.
- (b) After ordering on similarity and filtering on both similarity and importance, the star glyphs show structure



(a)



(b)

Idiom



System	DOSFA
What: Data How: Encode How: Facet How: Reduce	Table: many value attributes. Star plots. Small multiples with matrix alignment. Attribute filtering.



Aggregate

- Item Aggregation
- Spatial Aggregation
- Attribute Aggregation: Dimensionality Reduction

Aggregate



- In **aggregation**, a group of elements is represented by a new derived element that stands in for the entire group. Elements are merged together with aggregation
- Aggregation and filtering can be used in conjunction with each other.

Item Aggregation

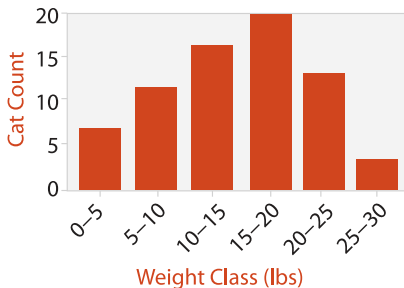


- The most straightforward use of **item aggregation** is within static visual encoding idioms; its full power and flexibility can be harnessed by interactive idioms where the view dynamically changes.



Example: Histograms

- The histogram idiom aggregates an arbitrary number of items into a concise representation of their distribution.



Idiom

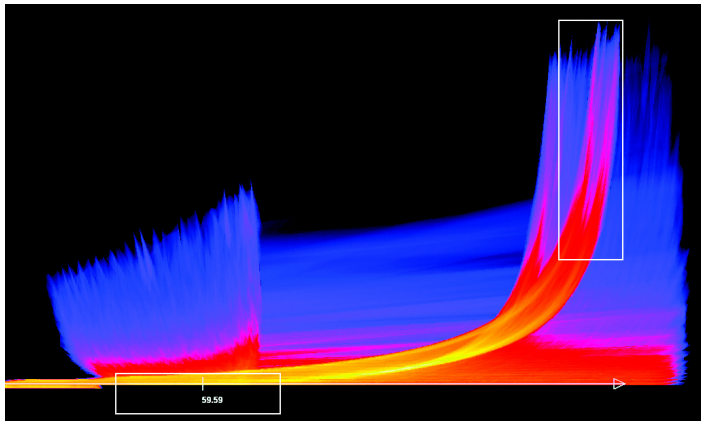


Idiom	Histograms
What: Data	Table: one quantitative value attribute.
What: Derived	Derived table: one derived ordered key attribute (bin), one derived quantitative value attribute (item count per bin).
How: Encode	Rectilinear layout. Line mark with aligned position to express derived value attribute. Position: key attribute



Example: Continuous Scatterplots

- The continuous scatterplot idiom uses color to show the density at each location, solving the problem of occlusion from overplotting and allowing scalability to large datasets



Idiom



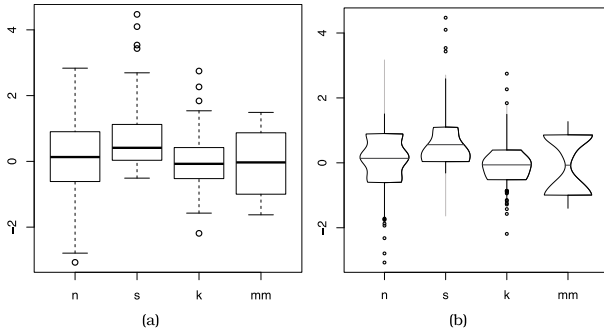
Idiom	Continuous Scatterplots
What: Data What: Derived	Table: two quantitative value attributes. Derived table: two ordered key attributes (x, y pixel locations), one quantitative attribute (overplot density).
How: Encode	Dense space-filling 2D matrix alignment, sequential categorical hue + ordered luminance colormap.
How: Reduce	Item aggregation.



Example: Boxplot Charts

The boxplot is an idiom presenting summary statistics for the distribution of a quantitative attribute, using five derived values. These plots illustrate four kinds of distributions: normal (n), skewed (s), peaked (k), and multimodal (mm).

- (a) Standard box plots.
- (b) Vase plots, which use horizontal spatial position to show density directly



Idiom



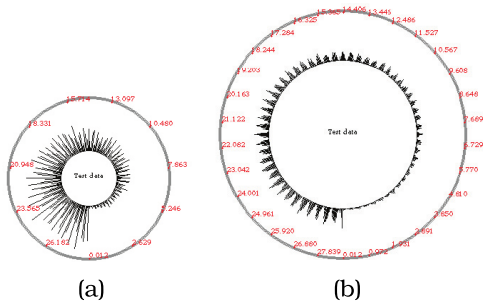
Idiom	Boxplot Charts
What: Data What: Derived Why: Tasks How: Encode How: Reduce Scale	Table: many quantitative value attributes. Five quantitative attributes for each original attribute, representing its distribution. Characterize distribution; find outliers, extremes, averages; identify skew. One glyph per original attribute expressing derived attribute values using vertical spatial position, with 1D list alignment of glyphs into separated with horizontal spatial position. Item aggregation. Items: unlimited. Attributes: dozens.



Example: SolarPlot

The SolarPlot circular histogram idiom provides indirect control of aggregation level by changing the circle size.

- (a) The small circle shows the increase in ticket sales over time.
- (b) Enlarging the circle shows seasonal patterns in addition to the gradual increase



Idiom



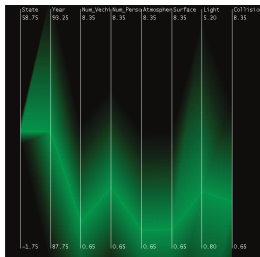
Idiom	SolarPlot
What: Data What: Derived	Table: one quantitative attribute. Derived table: one derived ordered key attribute (bin), one derived quantitative value attribute (item count per bin). Number of bins interactively controlled.
How: Encode	Radial layout, line marks. Line length: express derived value attribute; angle: key attribute.
How: Reduce Scale	Item aggregation. Original items: unlimited. Derived bins: proportional to screen space allocated



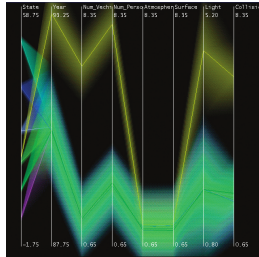
Example: Hierarchical Parallel Coordinates

Hierarchical parallel coordinates provide multiple levels of detail.

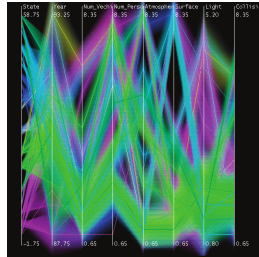
- (a) The single top cluster has large extent.
- (b) When several clusters are shown, each has a smaller extent.
- (c) When many clusters are shown, the proximity-based coloring helps them remain distinguishable from each other



(a)



(b)



(c)

Idiom

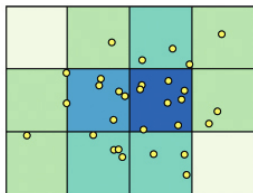


Idiom	Hierarchical Parallel Coordinates
What: Data What: Derived How: Encode How: Reduce Scale	Table. Cluster hierarchy atop original table of items. Five per-cluster attributes: count, mean, min, max, depth. Parallel coordinates. Color clusters by proximity in hierarchy. Interactive item aggregation to change level of detail. Items: 10,000–100,000. Clusters: one dozen.



Spatial Aggregation

- The challenge of **spatial aggregation** is to take the spatial nature of data into account correctly when aggregating it
- **Modifiable Areal Unit Problem (MAUP)** example, showing how different boundaries for aggregation regions lead to very different visual patterns on choropleth maps. (a) Central region is high density. (b) Central region is medium density. (c) Central region is low density



(a)



(b)

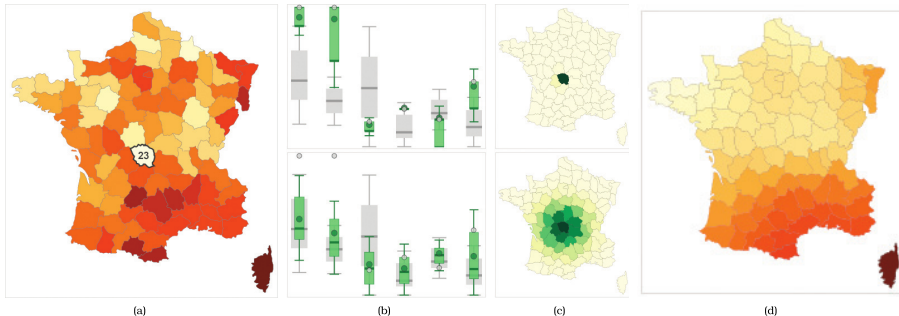


(c)



Example: Geographically Weighted Boxplots

- Geowigs are geographically weighted interactive graphics. (a) A choropleth map showing attribute x1. (b) The set of gw-boxplots for all six attributes at two scales. (c) Weighting maps showing the scales: local and larger. (d) A gw-mean map at the larger scale



Idiom



Idiom	Geographically Weighted Boxplots
What: Data	Geographic geometry with area boundaries. Table: Key attribute (area), several quantitative value attributes. Table: Five-number statistical summary distributions for each original attribute.
What: Derived	Multidimensional table: key attribute (area), key attribute (scale), quantitative value attributes (geographically weighted statistical summaries for each area at multiple scales).
How: Encode	Boxplot.
How: Facet	Superimposed layers: global boxplot as gray background, current-scale boxplot as green foreground.
How: Reduce	Spatial aggregation.

Attribute Aggregation: Dimensionality Reduction



- Just as attributes can be filtered, attributes can also be aggregated, where a new attribute is synthesized to take the place of multiple original attributes
- The goal of **dimensionality reduction** (DR), is to preserve the meaningful structure of a dataset while using fewer attributes to represent the items.



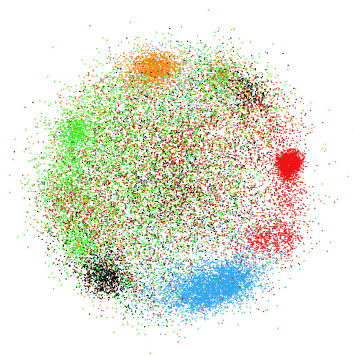
Why and When to Use DR?

- A small set of new synthetic attributes might be able to faithfully represent most of the structure or variance in the dataset.
- Nonlinear methods for dimensionality reduction are used when the new dimensions cannot be expressed in terms of a straightforward combination of the original ones.



Example: Dimensionality Reduction for Document Collections

- Dimensionality reduction of a large document collection using Glimmer for multidimensional scaling. The results are laid out in a single 2D scatterplot, allowing the user to verify that the conjectured clustering shown with color coding is partially supported by the spatial layout



Idiom

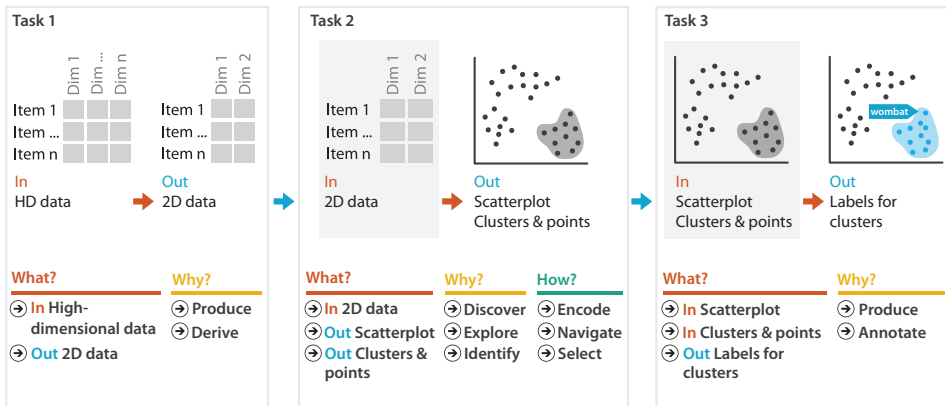


Idiom	Dimensionality Reduction for Document Collections
What: Data	Text document collection.
What: Derived	Table with 10,000 attributes.
What: Derived	Table with two attributes.
How: Encode	Scatterplot, colored by conjectured clustering.
How: Reduce	Attribute aggregation (dimensionality reduction) with MDS.
Scale	Original attributes: 10,000. Derived attributes: two. Items: 100,000.

Summary



- A chained sequence of what-why-how analysis instances for the scenario of dimensionality reduction of document collection data.

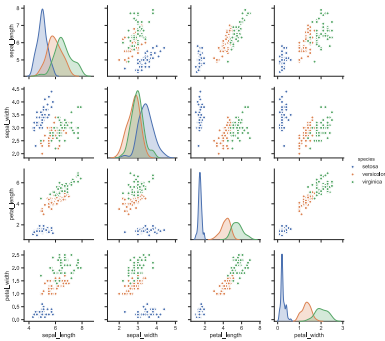


How to Show DR Data?



With standard dimensionality reduction techniques, the user chooses the number of synthetic attributes to create.

- When the target number of new attributes is two, the dimensionally reduced data is most often shown as a scatterplot.
- When more than two synthetic attributes are created, a scatterplot matrix (SPLOM) may be a good choice.



References



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